



# **Advanced Digital Image Processing Lecture #12**

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# Robotic Vision

- ❖ Robotic Vision: To perceive objects, manipulate objects and move in the environment
- ❖ Computer Vision: to perceive objects

# Introduction to Robot

- ❖ Robots: are physical agents that perform task by manipulating the physical world.
- ❖ They are equipped with effectors (arms, wheels, legs, etc.) and sensors (cameras, ultrasound measure the environment, accelerometers, etc.)
- ❖ Robotic systems need to embody prior knowledge about the robot, its physical environment, and the tasks that the robot will perform so that the robot can learn quickly and perform safely.



# Introduction to Robot (Categories of Robots)

- ❖ Manipulators: Fixed in work place. Uses as Industrial robots, surgical Assistant robots, etc.
- ❖ Mobile robot: Move about their environment using wheels, legs or similar mechanisms. Uses as nursing robots, cleaning robots, etc.
- ❖ Hybrid robot: Mobile robot equipped with manipulators. Example Humanoid robots. It includes artificial limbs, ears, and eyes for as like as humans.

# Examples of Robot

- ❖ Mobile robot
- ❖ Humanoid robot: Asimo, by honda, sony AI BO





# Examples of Robot



# Sensors

- ❖ Sensors: Sensors are the perceptual interface robots and their environments
- ❖ Active Sensors: send energy into the environment and rely on the fact that this energy is reflected back to the sensor.
- ❖ Passive Sensors: Are the true observers of the environment (Cameras)





# Sensors

- ❖ Range Finders: Many mobile robots make use of range finders, which are sensor that measure the distance to nearby objects
- ❖ Sonar Sensor: Sonar sensors emit directional sound waves, which are reflected by objects with some of the sound making it back into the sensor. The time and intensity of this returning signal thus carry information about the distance of nearby objects.



# Sensors

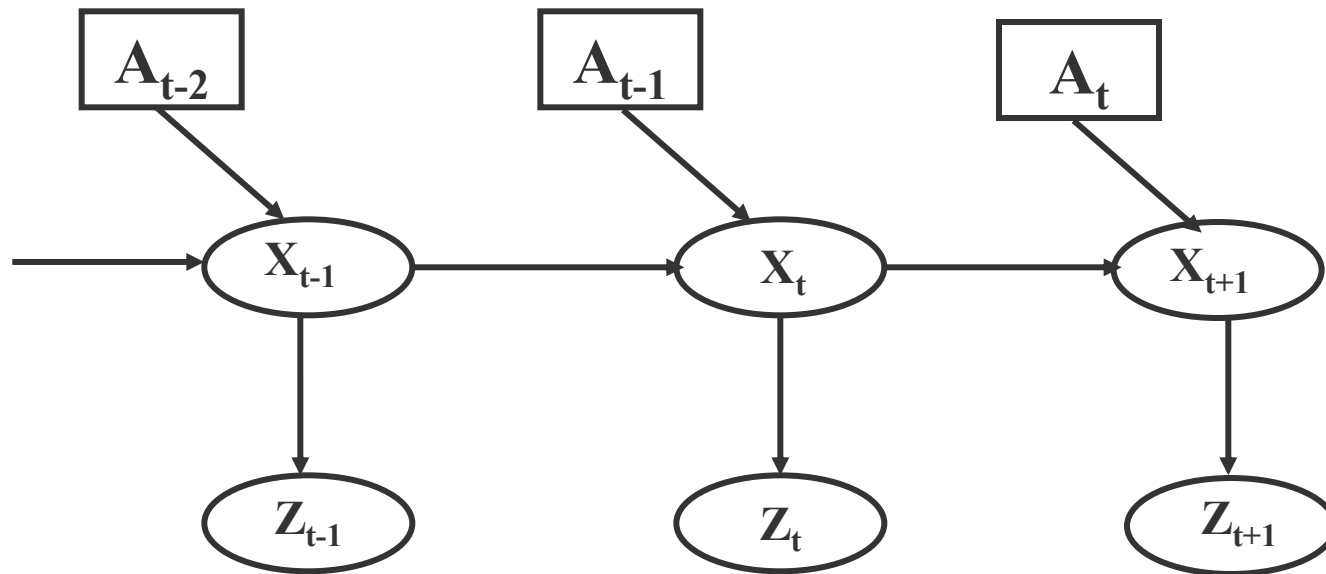
- ❖ Tactile Sensors: Close range sensors such as whiskers , touch-sensitive screen.
- ❖ Imaging Sensor: the cameras that provide us with images of the environment and using the computer vision techniques detect features of the environment. Stereo vision is particularly important in robotics.
- ❖ Proprioceptive sensors: Which inform the the robot of its own state.

# Robotic Perception

- ❖ Perception is the process by which robots map sensor measurement into internal representations of the environment.
- ❖ Robot perception can be viewed as temporal inference from sequences of actions and measurements as illustrated by this dynamic Bays network.

# Robotic Perception

- ❖  $X_t$  is the state of the environment (including robot) at time  $t$ ,  $Z_t$  is the observation received at time  $t$ ,  $A_t$  is the action taken after the observation is received.



# Localization & Tracking

- ❖ Localization is a generic example of robot perception. It is the problem of determining where things are.
- ❖ knowledge about where things are is at the core of any successful physical interaction of robot.
- ❖ Robot manipulators must know the location of objects they manipulate
- ❖ If the initial pose of the object to be localized is known, localization is a tracking problem.

# Mapping

- ❖ Localize of many objects is known as mapping.
- ❖ Imagine a robot that is not given a map of its environment, rather it has to construct such a map itself. This problem is known as mapping in robotics.
- ❖ The robot mapping problem is often referred to as simultaneous localization and mapping (SLAM)





# Planning to move

- ❖ The planning of robot motion is usually done in configuration space, where each point specifies the location and orientation of the robot and its joint angles
- ❖ The point to point motion problem is to deliver the robot or its end-effector to a designated target location.





# Robot Applications

- ❖ Industry & Agriculture
- ❖ Transportation
- ❖ Hazardous environment
- ❖ Exploration
- ❖ Health Care
- ❖ Personal Service
- ❖ Cleaning
- ❖ Entertainment
- ❖ Human Augmentation



## List of Reference Books

- ❖ Stuart Russell, Peter Norvig, “Artificial Intelligence, A modern Approach”
- ❖ Dana H. Ballard, “Computer Vision”
- ❖ David H. Hubel, “Eye, Brain, and Vision”
- ❖ Dwayne Phillips, “Image Processing in C”